

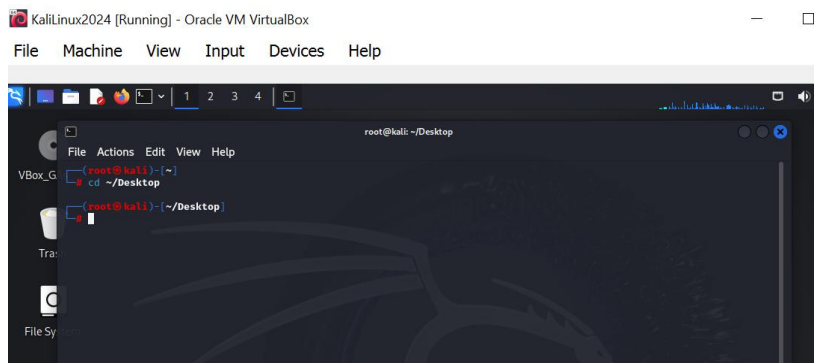
Password Cracking Lab Report

Step 1: Go to the Desktop Folder

Command:

cd ~/Desktop

This command opens the **Desktop folder** so we can save files there.



Step 2: Create Password Hashes

We are creating 3 password hashes:

- ❖ 1 Weak Password → abc123
- ❖ 2 Strong Passwords → Cybergenius28 and t0byD0g\$

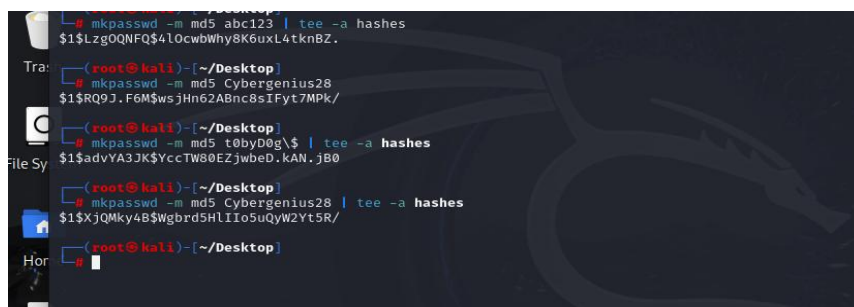
Commands:

mkpasswd -m md5 abc123 | tee -a hashes

mkpasswd -m md5 Cybergenius28 | tee -a hashes

mkpasswd -m md5 t0byD0g\$ | tee -a hashes

- mkpasswd creates an **MD5 hash** of the password
- -m md5 means use MD5 format
- tee -a hashes saves the hash into a file called hashes
- \\$ means we are escaping the \$ symbol so Linux understands it



Step 3: Check Hash File

Command:

cat hashes

This shows the contents of the hashes file (all 3 password hashes).

```
(root@kali) ~/Desktop
# cat hashes
$1$Lzg0QNFQ$4l0cwbWhy8K6uxL4tknBZ.
$1$advYA3JK$YccTW80EZjwbeD.kAN.jB0
$1$XjQMky4B$Wgbrd5HlIIo5uQyW2Yt5R/
(root@kali) ~/Desktop
#
```

Step 4: Use John the Ripper (Brute Force Attack)

Command:

john hashes --incremental

This tells **John the Ripper** to start guessing passwords **letter by letter** using brute force. It will try to crack all hashes.

- ✓ Press space to see progress.
- ✓ Press q to quit.

```
(root@kali) ~/Desktop
# john hashes --incremental
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 3 password hashes with 3 different salts (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
abc123 (?)
lg 0:00:00:06 0.1661g/s 23665p/s 49530c/s 49530C/s mich20..mistle
Use the "--show" option to display all of the cracked passwords reliably
Session aborted
(root@kali) ~/Desktop
#
```

Step 5: Show Cracked Passwords (John)

Command:

john hashes --show

This shows which passwords were cracked by John. You should see at least abc123 (the weak one).

```
(root@kali)~[~/Desktop]
# john hashes --show
?:abc123

1 password hash cracked, 2 left

(root@kali)~[~/Desktop]
#
```

Step 6: Identify Hash Type

Command:

hash-identifier

It tells you **what kind of hash it is** (like MD5, SHA1, etc). For this lab, the hash will show something like **"MD5(Unix)"**.

```

root@kali:~/Desktop# hash-identifier
#####
#
#          W G L L v1.2 #
#          By Zion3R #
#          www.Blackploit.com #
#          Root@Blackploit.com #
#####

HASH: $1$XjQMky4B$Wgbrd5HlIo5uQyW2Yt5R/

Possible Hashs:
[+] MD5(Unix)

HASH: █

```

Step 7: Unzip the Wordlist

Command:

```
sudo gzip -d /usr/share/wordlists/rockyou.txt.gz
```

Step 8: Use John the Ripper with Wordlist

Command:

john --wordlist=/usr/share/wordlists/rockyou.txt hashes

This tells John to use the rockyou.txt wordlist to crack the remaining strong passwords.



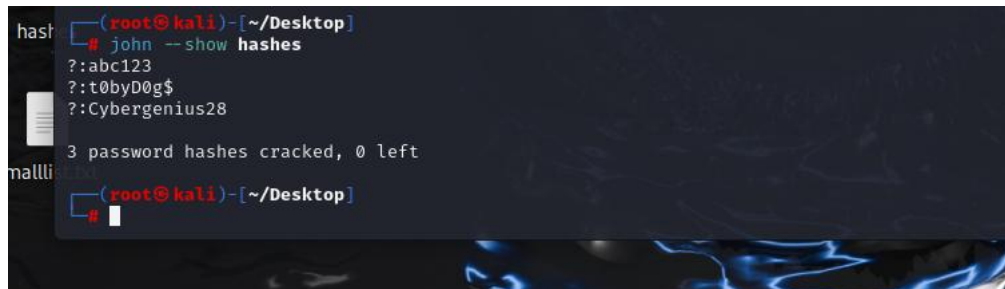
```
(root@kali)~[~/Desktop]
# john --wordlist=/usr/share/wordlists/rockyou.txt hashes
Warning: detected hash type "md5crypt", but the string is also recognized as "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading these as that type instead
Using default input encoding: UTF-8
Loaded 3 password hashes with 3 different salts (md5crypt, crypt(3) $1$ (and variants) [MD5 256/256 AVX2 8x3])
Remaining 2 password hashes with 2 different salts
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
Cybergenius28 (?)
t0byD0g$ (?)
2g 0:00:00:35 DONE (2025-05-31 08:50) 0.05560g/s 34476p/s 63813c/s 63813C/s t0yn00396..t.i.wifey
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Step 9: Show Cracked Passwords

Command:

john --show hashes

This shows all passwords John successfully cracked, including the strong ones.



```
(root@kali)~[~/Desktop]
# john --show hashes
?:abc123
?:t0byD0g$
?:Cybergenius28

3 password hashes cracked, 0 left
(root@kali)~[~/Desktop]
#
```